



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST -2, AUGUST - 2024

Time Allowed : 1.30 Hours]

MATHEMATICS

[Max. Marks : 45

Part - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer. 10x1=10
3. Each question carries 1 mark.

1. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies
 (1) $|k| \leq 6$ (2) $k = 0$ (3) $|k| > 6$ (4) $|k| \geq 6$
2. The number of positive zeros of the polynomial $\sum_{j=0}^n {}^nC_r (-1)^r x^r$ is
 (1) 0 (2) n (3) $< n$ (4) r
3. If $\sin^{-1} x + \sin^{-1} y = 2\pi/3$; then $\cos^{-1} x + \cos^{-1} y$ is equal to
 (1) $2\pi/3$ (2) $\pi/3$ (3) $\pi/6$ (4) π
4. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101} + y^{101} + z^{101}}$ is
 (1) 0 (2) 1 (3) 2 (4) 3
5. $\tan^{-1}(1/4) + \tan^{-1}(2/9)$ is equal to
 (1) $1/2 \tan^{-1}(3/5)$ (2) $1/2 \sin^{-1}(3/5)$ (3) $1/2 \tan^{-1}(3/5)$ (4) $\tan^{-1}(1/2)$
6. The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is
 (1) $4/3$ (2) $4/\sqrt{3}$ (3) $2\sqrt{3}$ (4) $3/2$
7. The centre of the circle inscribed in a square formed by the lines $x^2 - 8x - 12 = 0$ and $y^2 - 14y + 45 = 0$ is
 (1) (4, 7) (2) (7, 4) (3) (9, 4) (4) (4, 9)
8. The area of quadrilateral formed with foci of the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is
 (1) $4(a^2 + b^2)$ (2) $2(a^2 + b^2)$ (3) $a^2 + b^2$ (4) $1/2(a^2 + b^2)$
9. $\cos^{-1}(x) =$
 (1) $\cos^{-1} x$ (2) $-\cos^{-1} x$ (3) $\pi - \cos^{-1} x$ (4) $2\pi - \cos^{-1} x$
10. $3x^2 + 3y^2 - 4x + 3y + 10 = 0$ represent
 (1) Parabola (2) Ellipse (3) Hyperbola (4) None

PART - II

1. Answer any 4 questions. 2. Each question carries 2 marks

4x2=8

3. Question number 16 is compulsory

11. Solve the equation $3x^3 - 26x^2 + 52x - 24 = 0$ if its roots form an geometric progression.



12. Sketch the graph of $y = \sin\left(\frac{1}{3}x\right)$ for $0 \leq x < 6\pi$.

13. Find the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$

14. If $y = 4x+c$ is a tangent to the circle $x^2+y^2=9$, find c .

15. Find the equation of the hyperbola in each of the cases given below: foci $(\pm 2, 0)$, eccentricity $= 3/2$.

16. $y = x + c$ is a tangent to parabola $y^2 = 8x$, then find the value of c also find the point of contact.

PART - II

1. Answer any 4 questions. 2. Each question carries 3 marks

3. Question number 22 is compulsory

17. Find solution, if any, of the equation $2\cos^2 x - 9\cos x + 4 = 0$.

18. For what value of x , the inequality $\pi/2 < \cos^{-1}(3x - 1) < \pi$ holds?

19. Find the value of $\tan^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

20. Find the equation of circles that touch both the axes and pass through $(-4, -2)$ in general form.

21. Prove that the point of intersection of the tangents at t_1 and t_2 on the parabola $y^2 = 4ax$ is $[at_1 t_2, (at_1 + t_2)]$

22. Find the Domain of $\tan^{-1}(\sqrt{x^2 - 5x + 6})$.

PART - IV

1. Answer all the questions. 2. Each question carries 5 marks

23. a) Solve the equation $(x - 2)(x - 7)(x - 3)(x + 2) + 19 = 0$.

(OR)

b) Solve the equations (i) $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

24. a) Find the value of $\sin(\tan^{-1}(1/2) - \cos^{-1}(4/5))$

(OR)

b) If $a_1, a_2, a_3, \dots, a_n$ is an arithmetic progression with common difference d , prove that

$$\tan\left[\tan^{-1}\left(\frac{d}{1+a_1 a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2 a_3}\right) + \dots + \tan^{-1}\left(\frac{d}{1+a_{n-1} a_n}\right)\right] = \frac{a_n - a_1}{1 + a_1 a_n}$$

25. a) Find the vertices, foci for the hyperbola $9x^2 - 16y^2 = 144$.

(OR)

b) Assume that water issuing from the end of a horizontal pipe, 7.5 m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2.5 m below the line of the pipe, the flow of water has curved outward 3 m beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?